

Khawaja Yunus Ali University

School of Science and Engineering

Dept. of Computer Science and Engineering

Mid-term Examination: Fall-2019

Program: B.Sc. in CSE (Batch: 8th) 1st Year 2nd Semester

Course Code: PHY 1201

Course Title: Physics – II

Time: 1 hour & 30 minutes

Total Marks: 30

Group-B

[Answer any 5 (Five) questions including No.1]

5 × 5 = 25

1. See the table given below and answer the following questions:

Engine	Source		Sink	
	Temperature °C	Absorbed Heat (J)	Temperature °C	Rejected Heat (J)
Engine-I	327	X	27	600
Engine-II	527	1800	127	900

- a) Find the value of X. 2
- b) Which engine is better? – Justify your answer. 3
2. a) Explain the frequency postulate of Bohr atomic model. 2
- b) Draw the spectral series of hydrogen atom and calculate the wave number for Lyman series. 3
3. a) Calculate the work done in isochoric process. 2
- b) Investigate the basic differences between Ideal gas's and Van der Waal's equation. 3
4. a) Comments about the far future of universe related to entropy with logical explanation. 3
- b) Draw the pressure – volume curve for adiabatic and isothermal process. 2
5. a) What is degree of freedom in thermodynamics? 2
- b) The mean free path of the molecules of a gas is $6 \times 10^{-8}\text{m}$ and the diameter of a molecule is $2.5 \times 10^{-10}\text{m}$. Find the number of molecules per m^3 . 3
6. a) Define: (any three) 3
- i) Elastic collision
- ii) Mean free path
- iii) Mayer hypothesis
- iv) Entropy
- v) Thermometric substances
- c) Establish the relation between Enthalpy & Gibbs free energy. 2

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[Answer all questions]

Time: 10 min

0.5 × 10 = 5

[Tick (✓) mark to the appropriate one]

1. What is the unit of mean free path?

a) Jkg^{-1}

b) JK^{-1}

c) $Jkg^{-1}K^{-1}$

d) Js

e) m

2. Entropy remain constant in Process.

a) reversible

b) irreversible

c) adiabatic

d) isothermal

e) isobaric

3. What is the value of Plank constant?

a) 6.23×10^{23}

b) $1.6 \times 10^{-19} \text{Coul}$

c) $6.63 \times 10^{-34} \text{Js}$

d) $9.1 \times 10^{-31} \text{kg}$

e) $8.85 \times 10^{-12} \text{Fm}^{-1}$

4. The difference between two molar specific heats equal to

a) h (Plank constant)

b) R (molar gas constant)

c) γ (thermodynamical quantity)

d) Q (heat)

e) S (entropy)

5. Which one is only possible in slower process?

a) Adiabatic change

b) Isochoric change

c) Isobaric change

d) Isothermal change

e) Iso-entropic change

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- Examiner Signature**